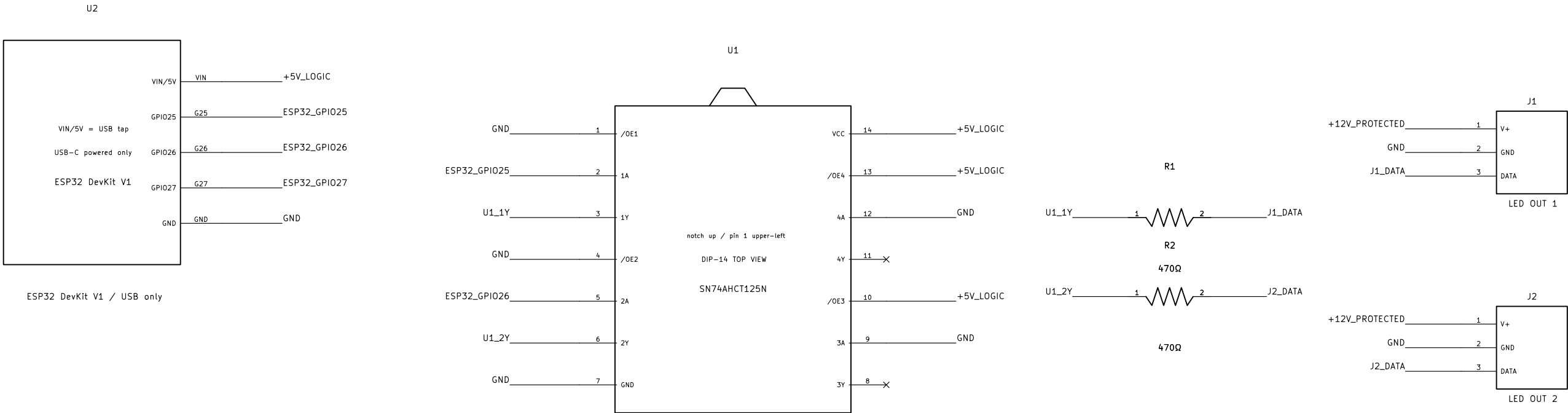


ESP32 LED Controller v6 – KiCad bench schematic

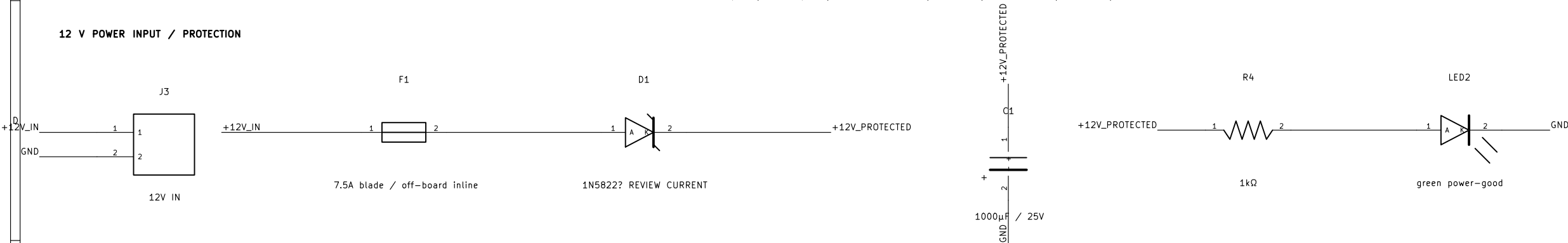
Rails: +12V_PROTECTED powers LED V+ only. +5V_LOGIC is USB-derived from ESP32 VIN/5V only. Grounds are common.



SN74AHCT125N / 74AHCT125 DIP-14

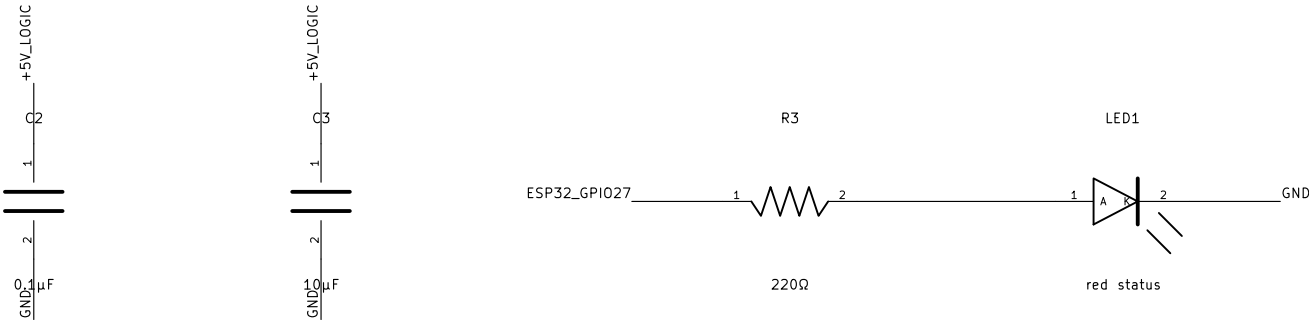
U1 unused channels: /OE3 pin10 and /OE4 pin13 → +5V_LOGIC; 3A pin9 and 4A pin12 → GND; 3Y pin8 and 4Y pin11 NC.

12 V POWER INPUT / PROTECTION



D1 current note: 1N5822 is not a free pass for multi-amp LED current. For full-current install, replace with rated protection or bypass if polarity is controlled.

5 V LOGIC / STATUS



Bench checks before power: +12V_PROTECTED ↔ +5V_LOGIC = open/megaohms; all GND points ≈0Ω; U1 pin14 gets +5V only from ESP32 USB/VIN; never backfeed ESP32 VIN from 12 V supply.